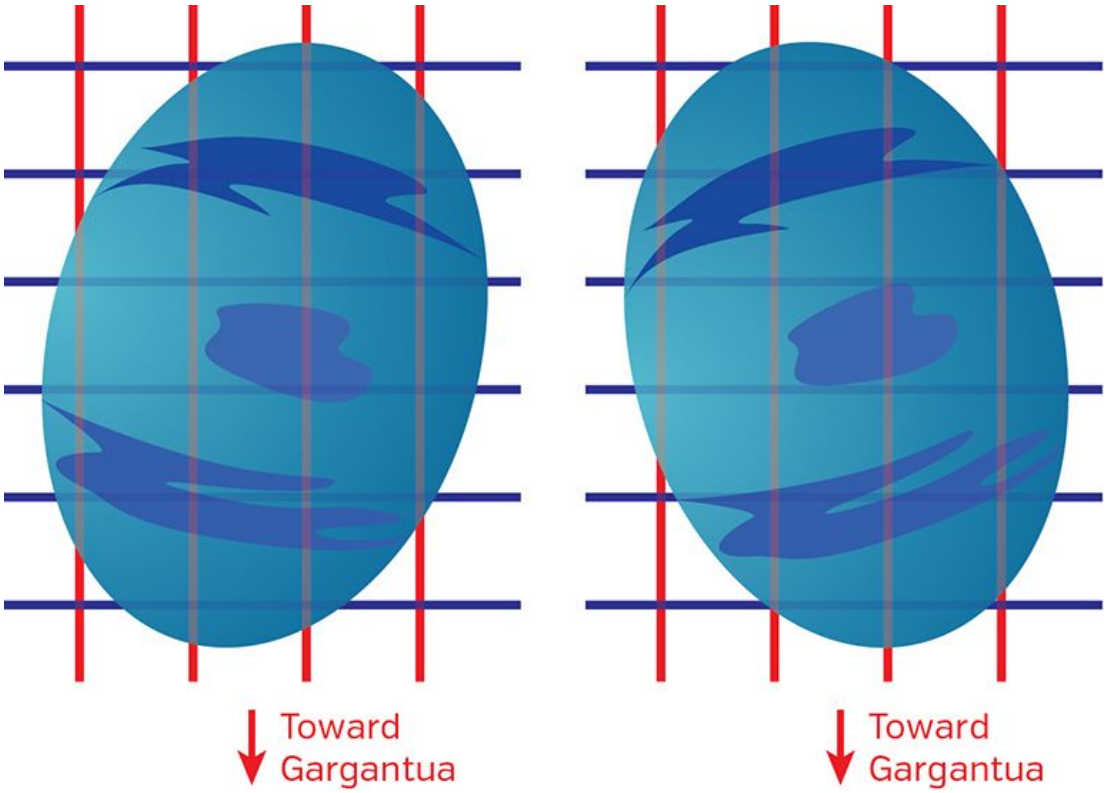


Fig. 17.6. The rocking of Miller's planet in response to Gargantua's tidal gravity: its stretching tendex lines (*red*) and squeezing tendex lines (*blue*).

The result is a simple rocking of the planet, back and forth, if the tilts are small enough that the planet's mantle isn't pulverized. When I computed the period of this rocking, how long it takes to swing from left to right and back again, I got a joyous answer. About an hour. The same as the observed time between giant waves, a time chosen by Chris without knowing about my science interpretation.

The first explanation for the giant waves, in my science interpretation, is a sloshing of the planet's oceans as the planet rocks under the influence of Gargantua's tidal gravity.

A similar sloshing, called "tidal bores," happens on Earth, on nearly flat rivers that empty into the sea. When the ocean tide rises, a wall of water can go rushing up the river; usually a tiny wall, but very occasionally respectably big. You can see an example in the top half of Figure 17.7: a tidal bore on the Qiantang River in Hangzhou, China, in